



Laboratory Report # 1

Name: Cyril Andre B. Durango

Date Completed: September 6, 2024

Laboratory Exercise Title: Design Flow of Digital Systems

Target Course Outcomes:

CO1: Create descriptions of digital hardware components such as in combinational and sequential circuits using synthesizable Verilog HDL constructs.

CO2: Verify the functionality of HDL-based components through design verification tools.

Exercise 1C: Half Adder Circuit. Create a new project following the detailed procedures in Exercise 1B and add this Verilog description of a half-adder circuit to the project. After the design entry, analyze and synthesize the design. After successful synthesis, use the RTL Viewer to check the schematic diagram of the synthesized design.

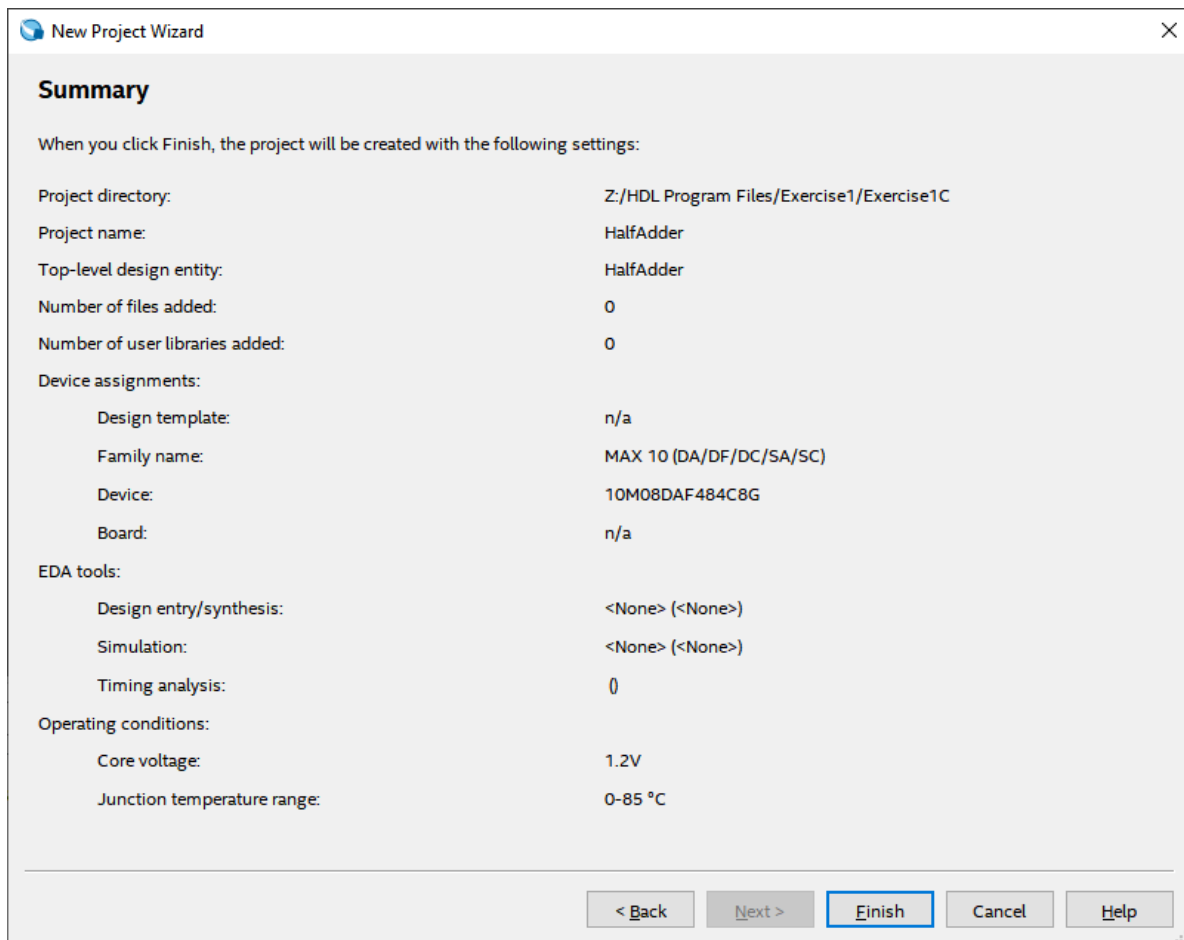


Figure 1. Project Settings Summary of the HalfAdder Project

A Half-adder is a logical circuit that adds two single-bit binary numbers, generating a sum and a carry output. Below is the truth table a half-adder circuit, Karnaugh maps of the outputs, and its circuit diagram:

Input		Output	
x	y	C	S
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

Table 1. Truth Table of a Half-Adder Circuit

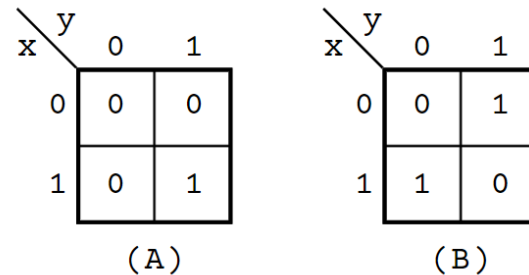


Figure 2. Karnaugh Maps for the Outputs. (A) K-map of the Carry or C-Output. (B) K-map of the Sum or S-Output

$$C = xy$$

$$S = x \oplus y$$

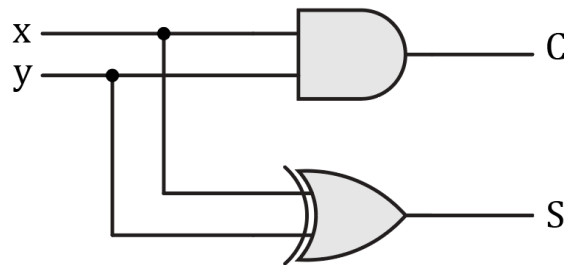


Figure 3. Half-adder Circuit Diagram

Using the equations that are derived from the K-maps, the gate primitives in Verilog can be utilized to simulate the logic behind the outputs.

```

1  /*****
2  * AUTHOR:      CYRIL ANDRE B. DURANGO
3  * CLASS:       CPE 3101L - INTRODUCTION TO HDL
4  * SCHEDULE:    Group 4 | FRIDAY | 7:30 AM - 10:30 AM
5  *
6  * PROJECT FILE TITLE: HalfAdder.v
7  * PROJECT DESCRIPTION:
8  *   A Verilog File of a Half Adder Circuit
9  *
10 /*****/
11
12 module HalfAdder (x, y, C, S);
13
14   input  x, y;
15   output C, S;
16
17   xor    x1(S, x, y);
18   and    A1(C, x, y);
19
20 endmodule |
  
```

Figure 4. Half-Adder.v Verilog File

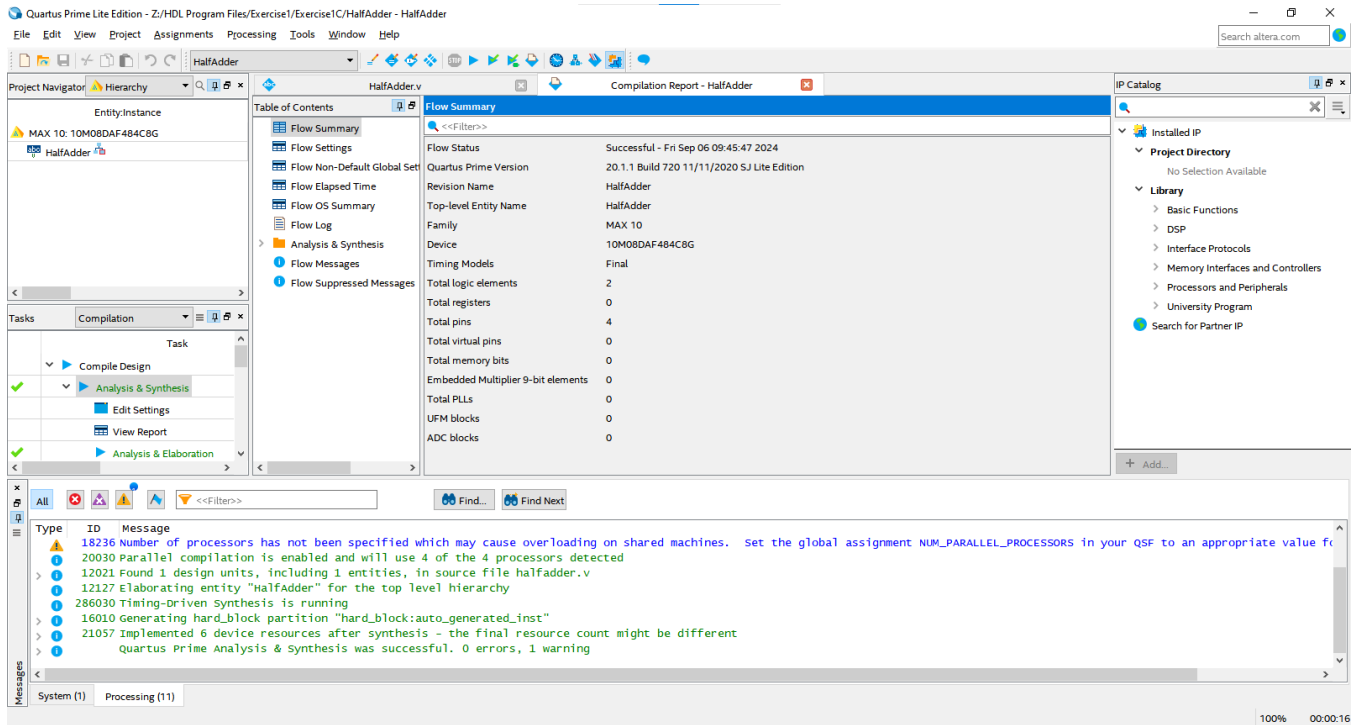


Figure 5. HalfAdder Verilog File Summary after Compilation

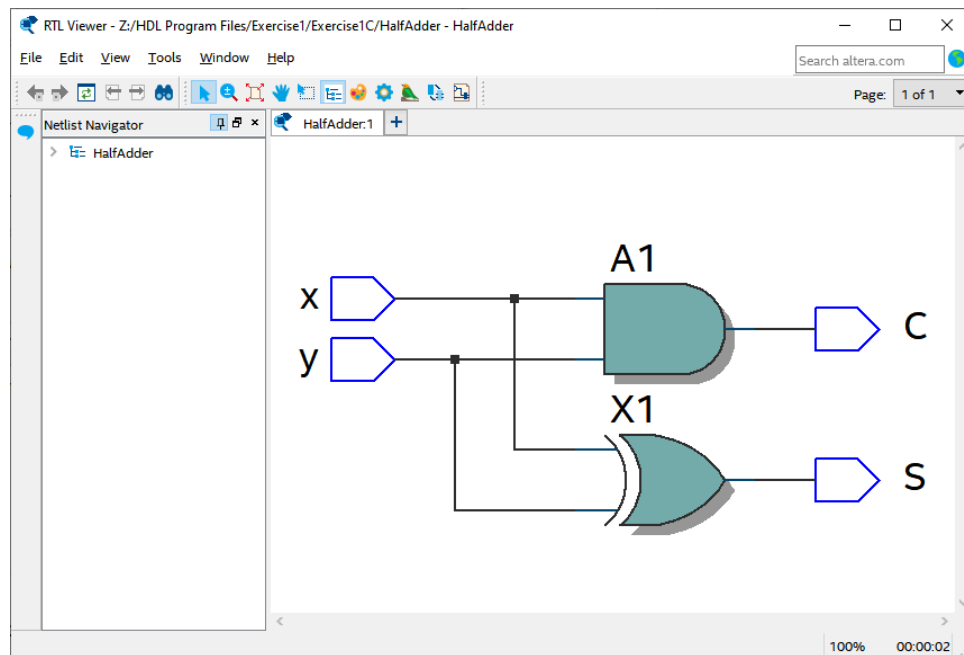


Figure 6. RTL Viewer of the Verilog file.

It can be seen that the synthesized circuit diagram in the RTL Viewer is the same as the logical circuit drawn in Figure 3. From this, it can be concluded that the Verilog file is coded similarly to how a half-adder circuit should work in the real world.